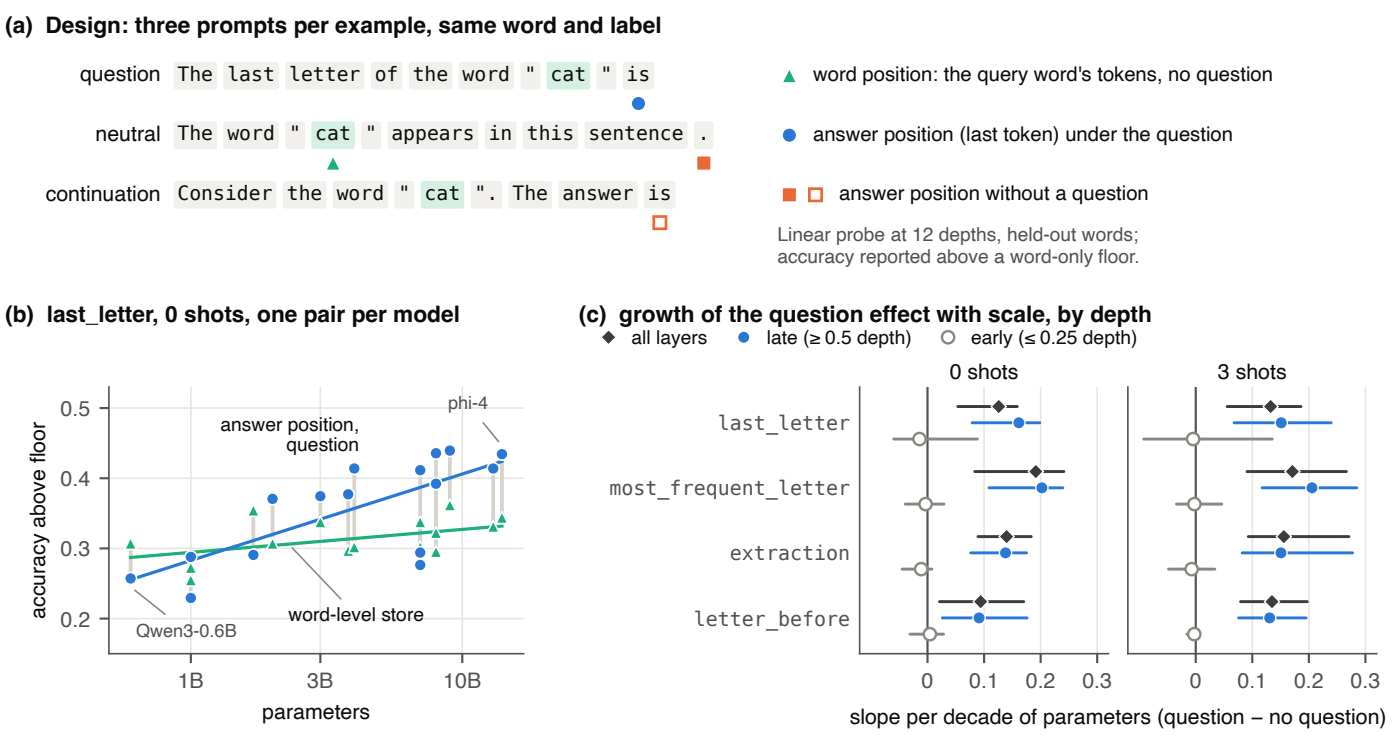
(a) Design: three prompts per example, same word and label
question The last letter of the word " cat " is
neutral The word " cat " appears in this sentence .
continuation Consider the word " cat ". The answer is
word position: the query word's tokens, no question
answer position (last token) under the question
answer position without a question
Linear probe at 12 depths, held-out words;
accuracy reported above a word-only floor.
(b) last_letter, 0 shots, one pair per model
accuracy above floor
0.5
0.4
0.3
0.2
answer position,
question
phi-4
word-level store
Qwen3-0.6B
1B
3B
10B
parameters
(c) growth of the question effect with scale, by depth
all layers
late (≥ 0.5 depth)
early (≤ 0.25 depth)
0 shots
3 shots
last_letter
most_frequent_letter
extraction
letter_before
0
0.1
0.2
0.3
slope per decade of parameters (question – no question)